Jingyang ZHAO

Visiting Scholar | Westlake University

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RESEARCH PROFILE

Visiting Scholar at Westlake University, working with Prof. Zibo Chen, with a Ph.D. in Materials Science and Engineering from the joint doctoral program of Zhejiang University and Westlake University. My research combines microfluidic platform engineering, quantitative biosensing, and soft biointerfaces. I design and fabricate microsystems, analyze transport and mechanics, and integrate experimental workflows for materials synthesis and biological measurement.

EDUCATION

Ph.D., Materials Science and Engineering 2022–2026

Zhejiang University / Westlake University, Hangzhou, China. DETECT Lab; advisor: Prof. Hanqing Jiang.

M.S., Mechanical Engineering 2021–2022

Columbia University, New York, USA. BioMEMS Lab; advisor: Prof. Qiao Lin.

Bachelor's degree 2016–2020

Zhejiang University, Hangzhou, China. Mechanical Design, Manufacturing and Automation; second major in English. Research advisor: Prof. Tao Zhang.

CAREER

Visiting Scholar, Westlake University 2027–present

Bioprogramming Lab, Hangzhou, China. Prof. Zibo Chen.

RESEARCH EXPERIENCE

Doctoral research: DETECT Lab, Westlake University, 2022–2026, with Prof. Hanqing Jiang. Earlier and continuing collaborations are identified below by institution, collaborator and original dates.

A. Microfluidic Systems and Automation

Automated microfluidic MOF synthesis [2]

DETECT Lab, Westlake University | Doctoral research

Established a microfluidic platform for controlled MOF synthesis, using ammonia manipulation and tunable reaction environments to regulate material formation and morphology.

Ammonia transport through PDMS

DETECT Lab, Westlake University | Submitted manuscript

Developed a separated donor–receiver approach to supply ammonia through PDMS without direct mixing, allowing independent control of ammonia delivery in microfluidic systems.

Integrated point-of-care diagnostics [3, 1]

DETECT Lab, Westlake University | Doctoral research

Investigated fluid–structure interaction and liquid handling for pneumatic liver-function testing; contributed to experimental investigation and analysis of a portable automated multiplexed qPCR platform. The studies address six liver-related biomarkers and fifteen HPV genotypes, respectively.

B. Biosensing and Biological Platforms

Synthetic-cell microfluidics

Westlake University | Collaboration with Prof. Zibo Chen since 2023

Designed microfluidic chips for high-throughput liposome/GUV generation (submitted). Fabricated 1 μm SU-8 mother-machine structures for tracking E. coli and GUVs under controlled conditions (ongoing).

Droplet platforms for biocatalyst screening

Westlake University | Prof. Longxing Cao | 2023–2026 | Submitted

Designed and fabricated droplet microfluidic platforms for encapsulating and manipulating reactions in large-scale screening of de novo enzyme libraries.

Microfluidic C. elegans assays

Westlake University | Collaboration with Prof. Kunliang Guan since 2024 | Ongoing

Developed immobilization and controlled-stimulation approaches for repeated-exposure experiments and quantification of learning- and memory-related behavioral responses.

Aptamer kinetics and intracellular mRNA assays [10, 9]

BioMEMS Lab, Columbia University | Prof. Qiao Lin | 2021–2022

Contributed to microfluidic assays for aptamer binding to proteins and cells, and molecular-beacon measurements of radiation-responsive intracellular BAX mRNA.

Aptamer-functionalized MOF FET sensing [8]

BioMEMS Lab, Columbia University | Prof. Qiao Lin | 2021–2022

Contributed to aptamer-functionalized MOF field-effect transistor biosensors for TNF-α detection.

Machine-learning analysis of digital immunoassays [12]

University of Michigan | Summer research visitor | Prof. Katsuo Kurabayashi | 2019

Developed convolutional-neural-network image analysis for high-density digital immunoassay microarrays, supporting multiplexed cytokine measurements.

C. Soft Biointerfaces and Robotics

Haptic interfaces and optogenetic probe mechanics [6, 7]

DETECT Lab, Westlake University | Doctoral collaborations

Fabricated miniature origami prototypes for a skin-conformable haptic interface. Analyzed tissue–probe mechanics in a separate collaboration on implantation-assistance-free flexible optogenetic waveguides.

Epidermal UV sensing [11]

DETECT Lab, Westlake University | Doctoral research

Contributed to material and device development for epidermal UV sensors with reversible, skin-color-specific visual readout.

Soft robotic fabrication and characterization [4, 5]

DETECT Lab, Westlake University | Doctoral collaborations

Used MEMS fabrication methods for millimeter-scale insect robots. Performed SEM characterization in a separate study of biodegradable origami robotic systems.

Flexible PCB interconnects

DETECT Lab, Westlake University | Submitted work

Designed connector interfaces for electrical and mechanical integration between transistor modules and control electronics.

Earlier training: Microfluidic chip fabrication, surface modification and on-chip experiments, Zhejiang University, 2018–2020, with Prof. Tao Zhang.

PUBLICATIONS

[1] J. Zhao, Y. Xiao, J. Li, C. Liang, H. Jiang. Discrimination for fifteen genotypes of human papillomavirus (HPV) via multiplexed qPCR in a portable and automated device. **Sensors and Actuators B: Chemical** 464, 140134 (2026). 10.1016/j.snb.2026.140134

[2] J. Zhao, C. Liang, J. Li, Y. Huang, C. Niu, H. Jiang. Continuous Synthesis of Optimized Ni3(HITP)2 With Broad Spectrum of Morphologies via Automated Microfluidic Ammonia Manipulation Platform. **Small Methods** 10, e70746 (2026). 10.1002/smtd.70746

[3] C. Liang, **J. Zhao**, P. Wei, K. Wei, H. Jiang. An energy-embodied paralleled liquid manipulation for equipment-free, quantitative multiplexed liver function monitoring. **Science Advances** 11, eadx0092 (2025). 10.1126/sciadv.adx0092

[4] C. Xu, Y. Cao, **J. Zhao**, Y. Cao, Y. Huang, Y. Lin, D. Wang, Z. Zhang, H. Jiang. Muscle-inspired elasto-electromagnetic mechanism in autonomous insect robots. **Nature Communications** 16, 6813 (2025). 10.1038/s41467-025-62182-2

[5] P. Wei, Z. Zhang, …, **J. Zhao**, …, L. Wang, H. Jiang. Biodegradable origami enables closed-loop sustainable robotic systems. **Science Advances** 11, eads0217 (2025). 10.1126/sciadv.ads0217

[6] M. T. Flavin, K. Ha, …, **J. Zhao**, …, H. Jiang, J. A. Rogers. Bioelastic state recovery for haptic sensory substitution. **Nature** 635, 345–352 (2024). 10.1038/s41586-024-08155-9

[7] Z. Chen, X. Li, …, **J. Zhao**, …, W. Gong, L. Li. Implantation-assistance-free flexible waveguide probe for optogenetic stimulation. **Cell Reports Physical Science** 5, 102217 (2024). 10.1016/j.xcrp.2024.102217

[8] W. Dai, K. Wen, X. Meng, S. Yu, **J. Zhao**, Q. Lin. Aptameric Metal–Organic Framework Nanobiosensor. **ACS Applied Nano Materials** 7, 17081–17091 (2024). 10.1021/acsanm.4c03305

[9] X. Meng, K. Wen, **J. Zhao**, Y. Han, S. A. Ghandhi, S. P. Kaur, D. J. Brenner, H. C. Turner, S. A. Amundson, Q. Lin. Microfluidic measurement of intracellular mRNA with a molecular beacon probe towards point-of-care radiation triage. **Sensors & Diagnostics** 3, 1344–1352 (2024). 10.1039/D4SD00079J

[10] K. Wen, X. Meng, C. Wang, **J. Zhao**, S. Botros, Q. Lin. Assessing aptamer-analyte binding kinetics by microfluidic fluorescence microscopy. **Sensors and Actuators B: Chemical** 401, 135018 (2024). 10.1016/j.snb.2023.135018

[11] Z. Yang, **J. Zhao**, C. Liang, H. Jiang. Materials and Device Design for Epidermal UV Sensors with Real-Time, Skin-Color Specific, and Naked-Eye Quasi-Quantitative Monitoring Capabilities. **Advanced Materials Technologies** 8, 2201481 (2023). 10.1002/admt.202201481

[12] Y. Song, **J. Zhao**, T. Cai, A. Stephens, S. Su, E. Sandford, C. Flora, B. H. Singer, M. Ghosh, S. W. Choi, M. Tewari, K. Kurabayashi. Machine learning-based cytokine microarray digital immunoassay analysis. **Biosensors and Bioelectronics** 180, 113088 (2021). 10.1016/j.bios.2021.113088

TECHNICAL SKILLS

Microfabrication: Soft lithography, photolithography, micromachining, surface modification and microfluidic device fabrication.

Simulation and design: COMSOL Multiphysics, fluid–structure interaction, mechanical analysis and microfluidic platform design.

Biological measurement: Aptamer assays, molecular-beacon measurements, droplet microfluidics, GUV generation and microfluidic cell manipulation.

Analysis and programming: Python, MATLAB, machine-learning-assisted image analysis; material and device characterization.

Languages: Chinese (native); English (professional).

TEACHING EXPERIENCE

Graduate Course Assistant, Columbia University

Materials/Processes in Manufacturing (MECE 3610), 2022; Advanced Manufacturing Processes (MECE 4610), 2021; Microelectromechanical Systems (MECE 4212), 2021.